GEETHANJALI INSTITUTE OF SCIENCE & TECHNOLOGY
(AN AUTONOMOUS INSTITUTION)
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QUESTION BANK (DESCRIPTIVE)

Subject Name with Code: Machine Learning (23A0529T)
Course & Branch: Year & Semester: III-II **Regulation: RG23**

UNIT – I

S. No.	Question	[BT Level] [CO][Marks]
2 Marks Questions (Short)		
1.	Define Machine Leaching.	L1, CO1,2M
2.	List the types of Machine Learning paradigms.	L1, CO1,2M
3.	List the modern types of Machine Learning	L1, CO1,2M
4.	List various applications of Machine Learning.	L1, CO1,2M
5.	Explain Learning by Rote	L2, CO1,2M
6.	Explain Learning by Induction.	L2, CO1,2M
7.	Explain Reinforcement Learning.	L2, CO1,2M
8.	Define Data and list various types of data.	L1, CO1,2M
9.	Explain Data Preprocessing.	L2, CO1,2M
10.	What are Outliers in data?	L1, CO1,2M
Descriptive Questions (Long)		
1	Define Machine Learning and explain its scope and applications	L2, CO1,10M
2	Explain the stages of Machine Learning with a neat diagram.	L2, CO1,10M
3	Explain the evolution of Machine Learning.	L2, CO1,10M
4	Differentiate Supervised, Unsupervised, and Reinforcement Learning	L2, CO1,10M
5	Discuss the types of data used in Machine Learning.	L2, CO1,10M
6	Solve the given data by standardization: a) A . **Student** / **Marks (X)**: A 50, B 60, C 70, D 80, E 90 b) E . **Employee** / **Age (years)** / **Salary (₹ in thousands)**: E1 22 25, E2 26 30, E3 30 35, E4 34 40, E5 38 45	L3, CO1,10M
7	Apply data normalization using range scaling and standardization for the following dataset: **Age (years)** / **Income (thousands of rupees)**: 25 400, 45 700, 65 1000	L3, CO1,10M

8	Explain the various sources of open-source datasets used in Machine Learning.	L2, CO1,10M
9	Explain Overfitting, Underfitting, and Good Fitting in Machine Learning	L2, CO1,10M
10	Explain common problems encountered with raw data	L2, CO1,10M

UNIT - II

S. No.	Question	[BT Level] [CO][Marks]
2 Marks Questions (Short)		
1.	Define Proximity Measures.	L1, CO2,2M
2.	What are the popular types of Proximity Measures?	L1, CO2,2M
3.	List the Distance Measures used in Machine Learning.	L1, CO2,2M
4.	Explain Radius Distance Nearest Neighbor.	L2, CO2,2M
5.	Explain Non-Metric Similarity Functions	L2, CO2,2M
6.	Explain Hamming Distance	L2, CO2,2M
7.	List the types of Proximity measures between binary patterns	L1, CO2,2M
8.	Explain the Confusion Matrix	L2, CO2,2M
9.	Define K-Nearest Neighbor (KNN) Classifier	L1, CO2,2M
10.	Define K-Nearest Neighbor (KNN) Regression	L1, CO2,2M
Descriptive Questions (Long)		
1	Explain the Distance Measures used in Machine Learning	L2, CO2,10M
2	Explain the K-Nearest Neighbor (KNN) Classifier algorithm	L2, CO2,10M
3	Explain the Radius Distance Nearest Neighbor algorithm	L2, CO2,10M
4	Explain the KNN Regression algorithm	L2, CO2,10M
5	Calculate the distance between players using KNN Classifier based on the following dataset and predict the class of sports for Kalyan (age 38, gender 0) with K=3	L3, CO2,10M

Name	Age	Gender	Class of Sports
Ajay	32	0	Football
Mark	40	0	Neither
Sara	16	1	Cricket
Zaira	34	1	Cricket
Sachin	55	0	Neither
Rahul	40	0	Cricket
Pooja	20	1	Neither
Smith	15	0	Cricket
Laxmi	55	1	Football
Michael	15	0	Football

6	A retail company wants to predict monthly sales revenue of new stores based on store size and number of employees using KNN Regression (K=3).	L3, CO2,10M

Store Size (sq ft)	Number of Employees	Monthly Sales Revenue ($)
1500	5	30,000
2000	7	50,000
2500	10	70,000
3000	15	80,000
3500	20	1,00,000

Predict revenue for stores of size 2800 and 4000 sq ft with 13 and 25

	employees.	
7	A bank wants to predict whether a customer is Eligible or Not Eligible for a loan based on Annual Income and Credit Score. **Customer** | **Annual Income (₹ lakhs)** | **Credit Score** | **Loan Status** C1 | 3 | 620 | Not Eligible C2 | 4 | 650 | Not Eligible C3 | 6 | 700 | Eligible C4 | 8 | 750 | Eligible C5 | 7 | 720 | Eligible Predict Loan Status of C6 (Annual Income = 5 lakhs, Credit Score = 680) using Radius Distance Nearest Neighbor (radius = 45, Euclidean distance).	L3, CO2,10M
8	Explain Proximity Measures between Binary Patterns	L2, CO2,10M
9	Explain common measurements used to evaluate the performance of classification algorithms	L2, CO2,10M
10	Explain common measurements used to evaluate the performance of Regression algorithms	L2, CO2,10M

UNIT - III

S. No.	Question	[BT Level] [CO][Marks]
2 Marks Questions (Short)		
1.	Explain Multi-Class Classification.	L2, CO3,2M
2.	List the commonly used loss functions in Regression Decision Trees.	L1, CO3,2M
3.	List any 4 properties of Decision Tree classifiers.	L1, CO3,2M
4.	Explain Information Gain	L2, CO3,2M
5.	Explain Misclassification Error.	L2, CO3,2M
6.	Explain Decision tree	L2, CO3,2M
7.	List any 3 impurity measures used in Decision Trees.	L1, CO3,2M
8.	Explain the Gini Index	L2, CO3,2M
9.	Define CART (Classification and Regression Tree)	L1, CO3,2M
10.	Define Bayes Rule.	L1, CO3,2M
Descriptive Questions (Long)		
1	Explain Impurity Measures used in Decision Trees	L2, CO3,10M
2	Apply the ID3 algorithm on the following dataset and construct a Decision Tree **Day** | **Outlook** | **Humidity** | **Wind** | **Play Tennis** D1 | Sunny | High | Weak | No D2 | Sunny | High | Strong | No D3 | Overcast | High | Weak | Yes D4 | Rain | High | Weak | Yes D5 | Rain | Normal | Weak | Yes	L3, CO3,10M

D6	Rain	Normal	Strong	No	
D7	Overcast	Normal	Strong	Yes	
D8	Sunny	High	Weak	No	
D9	Sunny	Normal	Weak	Yes	
D10	Rain	Normal	Weak	Yes	
D11	Sunny	Normal	Strong	Yes	
D12	Overcast	High	Strong	Yes	
D13	Overcast	Normal	Weak	Yes	
D14	Rain	High	Strong	No	

3	Using the following dataset, apply Naive Bayes Classifier to predict whether a new vehicle (Red, SUV, Domestic) is stolen:	L3, CO3,10M

No	Color	Type	Origin	Stolen
1	Red	Sports	Domestic	Yes
2	Red	Sports	Imported	Yes
3	Red	SUV	Imported	No
4	Yellow	Sports	Domestic	No
5	Yellow	SUV	Imported	Yes
6	Yellow	Sports	Domestic	Yes
7	Red	SUV	Imported	No

4	Explain the properties of Decision Tree classifiers.	L2, CO3,10M
5	Explain the Random Forest algorithm with a diagram.	L2, CO3,10M
6	Explain Decision Trees for Classification using a sample dataset.	L2, CO3,10M
7	A company wants to predict whether a person will buy a computer (Yes/No) based on Age, Income, and Credit Rating. Construct a Decision Tree classifier using Gini Index (CART algorithm):	L3, CO3,10M

Age	Income	Credit Rating	Buys Computer
Youth	High	Fair	No
Youth	High	Excellent	No
Middle	High	Fair	Yes
Senior	Medium	Fair	Yes
Senior	Low	Fair	Yes
Senior	Low	Excellent	No
Middle	Low	Excellent	Yes
Youth	Medium	Fair	No
Youth	Low	Fair	Yes
Senior	Medium	Fair	Yes

8	Explain Regression based on Decision Trees and the commonly used loss functions.	L2, CO3,10M
9	a) Explain Bias and Variance Trade-off b) Explain Bayes Rule and Bayesian Inference.	L2, CO3,10M
10	Explain Naive Bayes Classification (NBC) with its advantages and disadvantages	L2, CO3,10M

UNIT - IV

S. No.	Question	[BT Level] [CO][Marks]
2 Marks Questions (Short)		
1.	Define Linear Discriminant Analysis (LDA)	L1, CO4,2M
2.	Explain the Perceptron model	L2, CO4,2M
3.	List the types of kernels used in machine learning.	L1, CO4,2M

4.	Explain the concept of a Kernel	L2, CO4,2M
5.	Explain Simple Linear Regression	L2, CO4,2M
6.	Explain the concept of a Support Vector in SVM	L2, CO4,2M
7.	List the types of linear regression	L1, CO4,2M
8.	Explain Logistic Regression	L2, CO4,2M
9.	Define Support Vector Machine (SVM)	L1, CO4,2M
10.	Define the Optimum Hyperplane in SVM	L1, CO4,2M
Descriptive Questions (Long)		
1	Explain the Kernel Trick in Support Vector Machines (SVM)	L2, CO4,10M
2	Compute the Linear Discriminant (LDA) projection for the following 2D dataset Class A: $(1,2), (2,3), (3,3), (2,2)$ Class B: $(6,5), (7,7), (8,6), (7,5)$	L3, CO4,10M
3	Draw the hyperplane for the following data using a non-linear SVM: Positive labels: $(2,2), (10,6), (6,6), (6,10)$ Negative labels: $(1,1), (1,-1), (-1,-1), (-1,1)$	L3, CO4,10M
4	Calculate similarity measures between two vectors using Linear, Homogeneous Polynomial, Inhomogeneous Polynomial, and RBF Kernels. Use $x = \begin{bmatrix}3\\4\end{bmatrix}, y = \begin{bmatrix}4\\6\end{bmatrix}, C = 1$ and $\alpha = 1$.	L3, CO4,10M
5	Explain the difference between Linear Regression and Logistic Regress	L2, CO4,10M
6	a) Explain a Single-Layer Perceptron with a diagram b) Explain a Multi-Layer Perceptron (MLP) with a diagram	L2, CO4,10M
7	Explain Linear SVM and Nonlinear SVM	L2, CO4,10M
8	Explain the types of Kernel functions used in machine learning	L2, CO4,10M
9	a) Explain Linear Regression b) A company recorded the number of hours spent on online advertising and corresponding weekly sales (₹ lakhs): **Advertising Hours (x)** / **Weekly Sales (y) in ₹ lakhs** 2 / 10 4 / 15 6 / 19 8 / 25 10 / 30 12 / 36 Apply Linear Regression to predict weekly sales for Advertising Hours = 14 and 18.	L3, CO4,10M
10	a) Explain Logistic Regression. b) A bank wants to predict whether a customer will default on a loan (1) or not (0) based on annual income (₹ lakhs). The logistic regression model is: $z = -12 + 0.5x$ Using a threshold of 0.5, predict the probability of default for customers with income = 18 and 28 lakhs, and give the final prediction (0 or 1). Annual Income (x) / Default (1) / No Default (0) 10 / 1 15 / 1 20 / 0 25 / 0	L3, CO4,10M

30	0
35	0

UNIT - V

S. No.	Question	[BT Level] [CO][Marks]
2 Marks Questions (Short)		
1.	Explain Clustering.	L2, CO5,2M
2.	Explain Hard Clustering	L2, CO5,2M
3.	List the types of Hard Clustering.	L1, CO5,2M
4.	Explain Soft Clustering	L2, CO6,2M
5.	List the types of Soft Clustering	L1, CO6,2M
6.	Explain Agglomerative Clustering	L2, CO5,2M
7.	Explain Partitioning Clustering	L2, CO5,2M
8.	Explain the concept of a Dendrogram.	L2, CO5,2M
9.	Define Rough Clustering	L1, CO6,2M
10.	Define Euclidean Distance.	L1, CO6,2M
Descriptive Questions (Long)		
1	Explain the Matrix Factorization View of Clustering.	L2, CO5,10M
2	Given the dataset $X = \{A, B, C, D, E\}$and the following distance matrix, solve using Agglomerative Hierarchical Clustering A B C D E A 0 4 6 10 7 B 4 0 5 9 6 C 6 5 0 8 3 D 10 9 8 0 5 E 7 6 3 5 0	L3, CO5,10M
3	Use K-Means clustering to divide the dataset into 2 clusters. {1,2,6,8,3,7,10,12}	L3, CO5,10M
4	Use Fuzzy C-Means (FCM) Clustering to cluster the dataset $\{(1,3), (2,5), (6,8), (7,9)\}$	L3, CO6,10M
5	Explain Rough Clustering	L2, CO6,10M
6	Explain Soft Clustering algorithms	L2, CO6,10M
7	Discuss the advantages and disadvantages of K-Means clustering	L2, CO6,10M
8	Differentiate between Divisive and Agglomerative Clustering	L2, CO5,10M
9	Explain Hard Clustering algorithms	L2, CO5,10M
10	Given the dataset $Z = \{a, b, c, d, e\}$and the following distance matrix, apply Divisive Analysis Clustering Algorithm to form clusters: **a** **b** **c** **d** **e** **a** 0 9 3 6 11 **b** 9 0 7 5 10 **c** 3 7 0 9 2 **d** 6 5 9 0 8 **e** 11 10 2 8 0	L3, CO5,10M